

EMCH 201: Introduction to Application of Numerical Methods

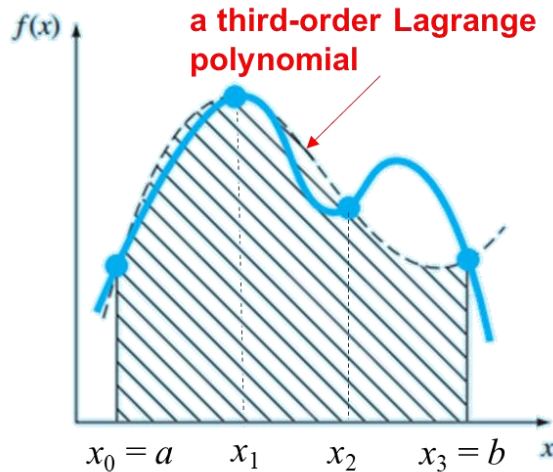
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Assignment 6

Assignment due date: **11:59 PM, Monday November 29th, 2021**

Point Possible: 100 (40+60)

1. With given four data points: $(x_0, f(x_0))$, $(x_1, f(x_1))$, $(x_2, f(x_2))$, and $(x_3, f(x_3))$, please derive the Simpson's 3/8 rule: $I = \int_a^b f(x)dx \cong \frac{3h}{8} [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)]$, where $h = (x_3 - x_0)/3$. Note that the four points are evenly distributed with spacing of h . (Hint, approximate $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3$, using similar approach that was used to derive Simpson's 1/3 rule. You can refer to Lecture 21) (40 Points)



1. With a given function: $f(x) = -5 + 2x - x^2 + 2x^3 - x^4$, please
 - 1) Integrate the function analytically: $\int f(x)dx$ (15 Points)
 - 2) Obtain the exact value of the integral between $x = 4$ and 10: $\int_4^{10} f(x)dx$ (15 Points)
 - 3) Use Simpson's 3/8 rule to perform the numerical integration between $x = 4$ and 10 (25 Points)
 - 4) Compute the true relative error ε for the application of Simpson's 3/8 rule above) (5 Points)